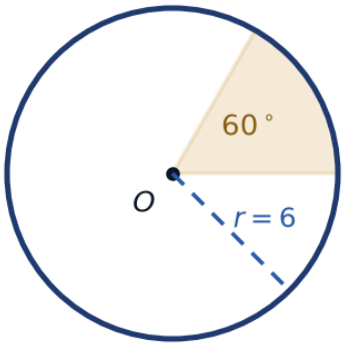
Functions of Several Variables & Partial Derivatives

ClipSAT Printable Practice Suite (clipsat.org) • Topic 1.01

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

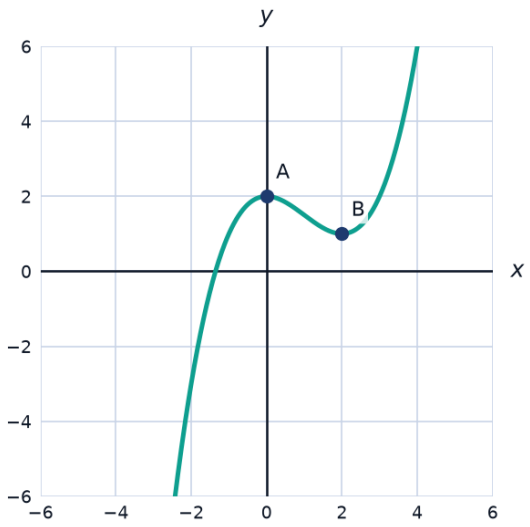
1 [Foundational] • Target: Domain, Range, and Level Curves

Verify Clairaut's Theorem on equality of mixed partial derivatives for the function $f(x, y) = \cos(2x - 3y) + x^2y^3$ by computing f_{xy} and f_{yx} directly.



2 [Foundational] • Target: First-Order Partial Derivatives

Let $f(x, y) = x^3e^{2y} - y\ln(x)$ defined on $x > 0$. Compute the first-order partial derivatives $f_x(x, y)$ and $f_y(x, y)$, and evaluate $f_x(1, 0)$ and $f_y(1, 0)$.



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3 [Foundational] • Target: Higher-Order Partials and Clairaut's Theorem

A temperature distribution is given by $T(x, y, z) = \frac{120}{1 + x^2 + 2y^2 + 3z^2}$. At the point $P(1, 1, 1)$, find: (a) the direction of steepest temperature increase, (b) the maximum rate of increase, and (c) the equation of the tangent plane to the isothermal level surface passing through P.

4 [Foundational] • Target: Tangent Planes and Linear Approximations

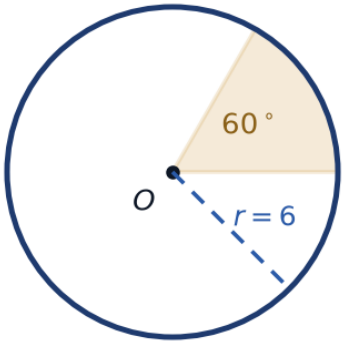
Evaluate the double integral of the function $f(x, y) = 2xy$ over the rectangular region $R = [0, 2] \times [0, 3]$ using Fubini's Theorem. Set up the iterated integral by integrating with respect to y first, then x.

5 [Foundational] • Target: Domain, Range, and Level Curves

Consider the vector field $\mathbf{F}(x, y) = \langle x^2, y \rangle$. According to Green's Theorem, the line integral of $\mathbf{F}$ around a simple closed curve C is equal to the double integral of $\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}$ over the region D enclosed by C, where $\mathbf{F} = \langle P, Q \rangle$. For the given vector field $\mathbf{F}(x, y) = \langle x^2, y \rangle$, identify the value of $\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}$.

6 [Foundational] • Target: First-Order Partial Derivatives

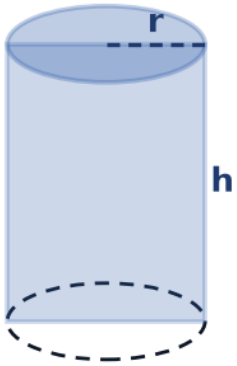
Let $z = f(x, y)$ be a differentiable function, with polar coordinates transformation $x = r\cos\theta$ and $y = r\sin\theta$. Express $\frac{\partial z}{\partial r}$ and $\frac{\partial z}{\partial \theta}$ in terms of $\frac{\partial z}{\partial x}$ and $\frac{\partial z}{\partial y}$, and prove that: $\left(\frac{\partial z}{\partial r}\right)^2 + \frac{1}{r^2}\left(\frac{\partial z}{\partial \theta}\right)^2 = \left(\frac{\partial z}{\partial x}\right)^2 + \left(\frac{\partial z}{\partial y}\right)^2 = ||ablaz||^2$.



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7 [Foundational] • Target: Higher-Order Partials and Clairaut's Theorem

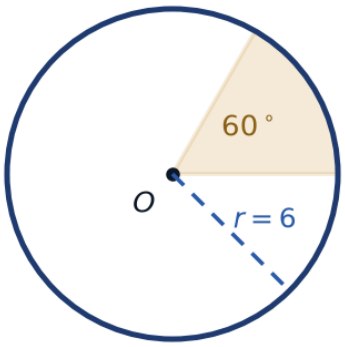
Use the method of Lagrange multipliers to find the maximum volume of a rectangular box with faces parallel to the coordinate planes inscribed in the ellipsoid $\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$.



Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Tangent Planes and Linear Approximations

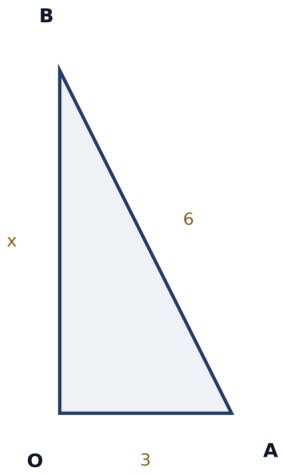
Derive the Gaussian integral $I = \int_{-\infty}^{\infty} e^{-x^2/2} dx = \sqrt{2\pi}$ from first principles by evaluating the double integral $I^2 = \iint_{\mathbb{R}^2} e^{-(x^2+y^2)/2} dA$ in polar coordinates.



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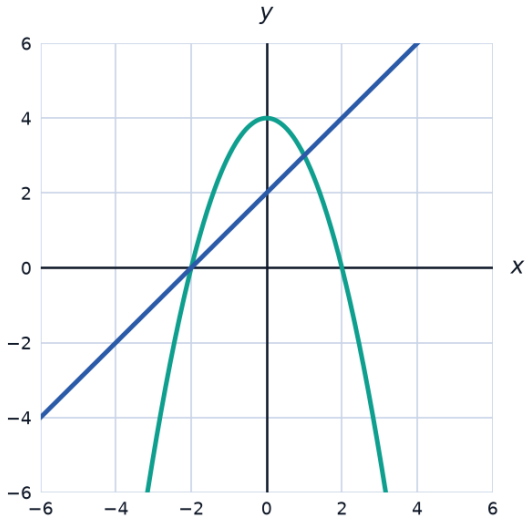
9 **[Intermediate]** • **Target: Domain, Range, and Level Curves**

Consider the region R in the first quadrant bounded by the x-axis, the y-axis, and the line $y = -2x + 6$. We are interested in evaluating the double integral $\iint_R f(x, y)\,dA$ where $f(x, y) = x + y$. However, the integral is set up in polar coordinates as $\int_{\alpha}^{\beta} \int_{r_1(\theta)}^{r_2(\theta)} f(r\cos\theta, r\sin\theta) \cdot r\,dr\,d\theta$. If this integral is evaluated from $\theta = 0$ to $\theta = \frac{\pi}{2}$, and the inner integral bounds are $r_1(\theta) = 0$ and $r_2(\theta) = \frac{6}{\sin\theta + 2\cos\theta}$, and the outer integral is $\int_0^{\pi/2} \int_0^{\frac{6}{\sin\theta + 2\cos\theta}} (r\cos\theta + r\sin\theta) \cdot r\,dr\,d\theta$, determine the value of α and β that correspond to the description of region R in polar coordinates for $0 \le \theta \le \frac{\pi}{2}$.



10 **[Intermediate]** • **Target: First-Order Partial Derivatives**

Evaluate the non-elementary integral $\int_0^1 \int_y^1 e^{x^2} dx\,dy$ by reversing the order of integration.



11 **[Intermediate]** • **Target: Higher-Order Partials and Clairaut's Theorem**

Use Stokes' Theorem to evaluate $\oint_C \mathbf{F} \cdot d\mathbf{r}$, where $\mathbf{F} = \langle -y^3, x^3, -z^3 \rangle$ and C is the intersection of the cylinder $x^2 + y^2 = 1$ and the plane $z = 1$, oriented counterclockwise when viewed from above. Verify surface independence by comparing the flux of $abla \times \mathbf{F}$ through the flat disk $S_1 : x^2 + y^2 \le 1, z = 1$ and the paraboloid cap $S_2 : z = 2 - (x^2 + y^2)$.

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12 [Intermediate] • Target: Tangent Planes and Linear Approximations

A particle's position in three-dimensional space is given by the vector function $\mathbf{r}(t) = \langle x(t), y(t), z(t) \rangle$, where t represents time. The velocity of the particle is given by $\mathbf{v}(t) = \mathbf{r}'(t)$. Suppose the velocity components are related by the partial differential equations: $\frac{\partial v_x}{\partial t} = -2x(t) + 3y(t)$ and $\frac{\partial v_y}{\partial t} = x(t) - 4y(t)$. If at time $t = 0$, the particle is at the origin $(0, 0, 0)$ and its velocity is $\mathbf{v}(0) = \langle 5, -2, 0 \rangle$, determine the value of $\frac{\partial^2 v_x}{\partial t^2}$ at $t = 0$.

13 [Intermediate] • Target: Domain, Range, and Level Curves

Use the Divergence Theorem to calculate the outward flux of $\mathbf{F} = \langle x^3 + y\sin z,\ y^3 + z\cos x,\ z^3 + xy \rangle$ across the closed boundary surface of the solid cylinder $E : x^2 + y^2 \le a^2,\ 0 \le z \le h$.

14 [Intermediate] • Target: First-Order Partial Derivatives

Use the Green's Theorem line integral formula for area, $A = \frac{1}{2}\oint_C (x\,dy - y\,dx)$, to derive the area enclosed by the hypocycloid (astroid) $\mathbf{r}(t) = \langle a\cos^3 t,\ a\sin^3 t \rangle$ for $0 \le t \le 2\pi$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Higher-Order Partials and Clairaut's Theorem • Cross-Lesson: 4.01 Double Integrals

Let the function $f(x, y)$ be defined as $f(x, y) = 2x^2 - 3y^2$. Calculate the partial derivative of f with respect to y at the point $(2, 3)$.

16 [Challenging] • Target: Tangent Planes and Linear Approximations • Cross-Lesson: 2.01 The Chain Rule & Directional Derivatives

Consider the vector field $\mathbf{F}(x, y, z) = \langle x, y, z \rangle$. Let S be the closed surface of the unit cube defined by $0 \le x \le 1, 0 \le y \le 1, 0 \le z \le 1$, oriented outwards. According to the Divergence Theorem, the flux of $\mathbf{F}$ through S is equal to the triple integral of the divergence of $\mathbf{F}$ over the volume V enclosed by S. Calculate this flux by evaluating $\iiint_V abla \cdot \mathbf{F}\, dV$.

17 [Challenging] • Target: Domain, Range, and Level Curves • Cross-Lesson: 3.01 Optimization: Critical Points & Lagrange Multipliers

Use Green's Theorem to evaluate $\oint_C (y^2 - 7y)\,dx + (2xy + 2x)\,dy$, where C is the positively oriented boundary of the region D enclosed between the parabolas $y = x^2$ and $x = y^2$.

18 [Challenging] • Target: First-Order Partial Derivatives • Cross-Lesson: 4.01 Double Integrals

Use Green's Theorem to evaluate the circulation $\oint_C (3y - e^{\sin x})\,dx + (7x + \sqrt{y^4 + 1})\,dy$, where C is the circle $x^2 + y^2 = 9$ oriented counterclockwise.

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19 [Challenging] • Target: Higher-Order Partials and Clairaut's Theorem • Cross-Lesson: 2.01 The Chain Rule & Directional Derivatives

Consider the function $f(x, y) = 3x^2 - 5y^3$. Find the partial derivative of f with respect to y evaluated at the point $(2, -1)$, i.e., $f_y(2, -1)$.

20 [Challenging] • Target: Tangent Planes and Linear Approximations • Cross-Lesson: 3.01 Optimization: Critical Points & Lagrange Multipliers

Find the flux $\iint_S \mathbf{F} \cdot d\mathbf{S}$ of $\mathbf{F} = \langle x, y, z \rangle$ across the part of the plane $x + y + z = 3$ in the first octant, with upward normal.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Domain, Range, and Level Curves

Find the domain of the function $f(x, y) = \ln(9 - x^2 - y^2)$.

a. The interior of the circle $x^2 + y^2 < 9$

b. The exterior of the circle $x^2 + y^2 > 9$

c. The closed disk $x^2 + y^2 \leq 9$

d. All of $\mathbb{R}^2$

22 [Foundational] • Target: Domain, Range, and Level Curves

What geometric curves are the level curves of $z = x^2 + y^2$ for $c > 0$?

a. Concentric circles centered at $(0, 0)$

b. Lines of slope 1

c. Parabolas opening upward

d. Hyperbolas

23 [Foundational] • Target: Domain, Range, and Level Curves

What is the range of $g(x, y) = \sqrt{16 - x^2 - y^2}$?

a. $[0, 4]$

b. $[0, 16]$

c. $[-4, 4]$

d. $[0, \infty)$

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24 [Foundational] • Target: First-Order Partial Derivatives

Find the partial derivative $f_x(x, y)$ for $f(x, y) = 3x^3y^2 - 5x^2 + 4y^3$.

a. $9x^2y^2 - 10x$

b. $6x^3y + 12y^2$

c. $9x^2y^2 - 10x + 12y^2$

d. $3x^2y^2 - 10x$

25 [Foundational] • Target: First-Order Partial Derivatives

Find $f_y(x, y)$ for $f(x, y) = e^{2x}\sin(3y)$.

a. $3e^{2x}\cos(3y)$

b. $2e^{2x}\sin(3y)$

c. $6e^{2x}\cos(3y)$

d. $-3e^{2x}\cos(3y)$

26 [Foundational] • Target: First-Order Partial Derivatives

Evaluate $f_x(1, 2)$ for $f(x, y) = x^2y + 3xy^2$.

a. 16

b. 28

c. 12

d. 8

27 [Foundational] • Target: Higher-Order Partials and Clairaut's Theorem

What does Clairaut's Theorem state about mixed second-order partial derivatives f_{xy} and f_{yx}?

a. If both f_{xy} and f_{yx} are continuous on an open region, then $f_{xy} = f_{yx}$

b. Mixed partials are always equal to 0

c. The sum $f_{xy} + f_{yx} = 0$

d. Equality holds only for polynomials

28 [Foundational] • Target: Higher-Order Partials and Clairaut's Theorem

For $f(x, y) = x^3y^4$, verify Clairaut's Theorem by finding f_{xy}.

a. $12x^2y^3$

b. $3x^2y^4$

c. $4x^3y^3$

d. $6xy^2$

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Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 **[Intermediate]** • **Target: Higher-Order Partials and Clairaut's Theorem**

Find f_{xx} for $f(x, y) = x^4 - 2x^2y^3 + y^5$.

a. $12x^2 - 4y^3$

b. $4x^3 - 4xy^3$

c. $12x^2$

d. $-12x^2y^2$

30 **[Intermediate]** • **Target: Tangent Planes and Linear Approximations**

What is the equation of the tangent plane to the surface $z = f(x, y)$ at the point (x_0, y_0, z_0)?

a. $z - z_0 = f_x(x_0, y_0)(x - x_0) + f_y(x_0, y_0)(y - y_0)$

b. $z - z_0 = f_x(x - x_0) - f_y(y - y_0)$

c. $z = f_x x_0 + f_y y_0$

d. $(x - x_0) + (y - y_0) + (z - z_0) = 0$

31 **[Intermediate]** • **Target: Tangent Planes and Linear Approximations**

Find the equation of the tangent plane to $z = x^2 + y^2$ at $(1, 2, 5)$.

a. $z - 5 = 2(x - 1) + 4(y - 2) \Leftrightarrow 2x + 4y - z = 5$

b. $2x + 4y + z = 15$

c. $x + 2y - z = 0$

d. $4x + 2y - z = 5$

32 **[Intermediate]** • **Target: Tangent Planes and Linear Approximations**

What is a normal vector to the surface $z = f(x, y)$ at (x_0, y_0)?

a. $\mathbf{n} = [f_x(x_0, y_0), f_y(x_0, y_0), -1]$

b. $\mathbf{n} = [f_x, f_y, 1]$

c. $\mathbf{n} = [1, 1, 1]$

d. $\mathbf{n} = [-f_y, f_x, 0]$

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33 **[Intermediate]** • **Target: Domain, Range, and Level Curves**

Consider the function $f(x, y) = 5x^3 - 2y^2 + 7$. Compute the value of the partial derivative $f_y(2, -3)$.

a. 20

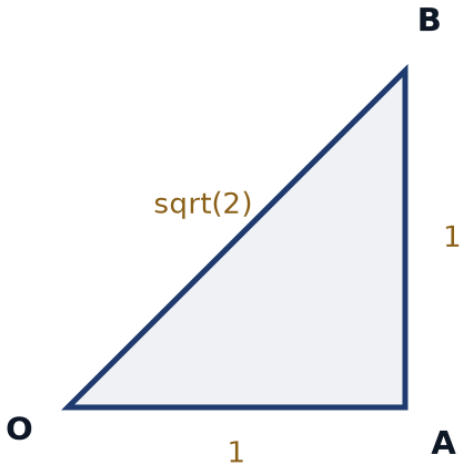
b. -12

c. 12

d. -20

34 **[Intermediate]** • **Target: First-Order Partial Derivatives**

Evaluate the double integral $\iint_R xy\,dA$ where R is the region in the first quadrant bounded by the lines $y = x$, $x = 1$, and $y = 0$.



a. 1/12

b. 1/8

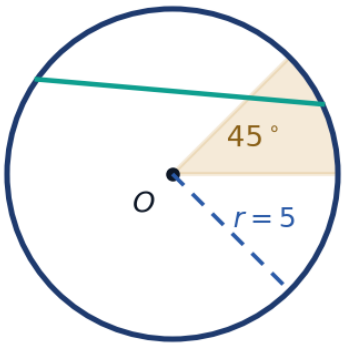
c. 1/6

d. 1/4

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35 [Intermediate] • Target: Higher-Order Partials and Clairaut's Theorem

Consider the region R in the first quadrant bounded by the x-axis, the line $y = x$, and the circle $x^2 + y^2 = 16$. The integral $\iint_R f(x, y)\, dA$ can be evaluated more easily in polar coordinates. If $f(x, y) = \frac{1}{\sqrt{x^2 + y^2}}$, and the integral in polar coordinates is given by $\int_0^{\pi/4} \int_0^4 r \cdot \frac{1}{r}\, dr\, d\theta = \int_0^{\pi/4} A\, d\theta$, find the value of A.



a. 4

b. $\pi/4$

c. 16

d. 2

36 [Intermediate] • Target: Tangent Planes and Linear Approximations

For $f(x, y) = 3x^2 + 4y^2$, find $f_x(4, 5)$.

a. 24

b. 30

c. 12

d. 40

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37 **[Intermediate]** • **Target: Domain, Range, and Level Curves**

For $f(x, y) = 7x^2 + 2y^2$, find $f_y(3, 6)$.



a. 24

b. 12

c. 42

d. -24

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

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38 **[Challenging]** • **Target: First-Order Partial Derivatives** • **Cross-Lesson: 4.01 Double Integrals**

For $f(x, y) = 7x^2 + 4y^2$, find $f_y(-3, 1)$.



a. -24

b. 4

c. 8

d. -42

39 **[Challenging]** • **Target: Higher-Order Partials and Clairaut's Theorem** • **Cross-Lesson: 2.01 The Chain Rule & Directional Derivatives**

For $f(x, y) = 8x^2 + 9y^2$, find $f_x(-4, -3)$.

a. -48

b. -64

c. -32

d. -54

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40 [Challenging] • Target: Tangent Planes and Linear Approximations • Cross-Lesson: 3.01 Optimization: Critical Points & Lagrange Multipliers

For $f(x, y) = 4x^2 + 7y^2$, find $f_y(-2, -6)$.

a. -28

b. -42

c. -84

d. -16

41 [Challenging] • Target: Domain, Range, and Level Curves • Cross-Lesson: 4.01 Double Integrals

For $f(x, y) = 9x^2 + 8y^2$, find $f_y(3, -5)$.

a. -80

b. 48

c. -40

d. 54

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42 [Challenging] • Target: First-Order Partial Derivatives • Cross-Lesson: 2.01 The Chain Rule & Directional Derivatives

For $f(x, y) = 8x^2 + 7y^2$, find $f_x(-4, 2)$.



a. 32

b. -32

c. 28

d. -64

43 [Challenging] • Target: Higher-Order Partials and Clairaut's Theorem • Cross-Lesson: 3.01 Optimization: Critical Points & Lagrange Multipliers

For $f(x, y) = 5x^2 + 6y^2$, find $f_x(-5, 1)$.



a. -50

b. 10

c. -25

d. 12

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44 [Challenging] • Target: Tangent Planes and Linear Approximations • Cross-Lesson: 4.01 Double Integrals

For $f(x, y) = 4x^2 + 7y^2$, find $f_x(-5, -2)$.



a. -16

b. -20

c. -28

d. -40

45 [Challenging] • Target: Domain, Range, and Level Curves • Cross-Lesson: 2.01 The Chain Rule & Directional Derivatives

For $f(x, y) = 4x^2 + 3y^2$, find $f_y(-4, -3)$.



a. -18

b. -24

c. -9

d. -32

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